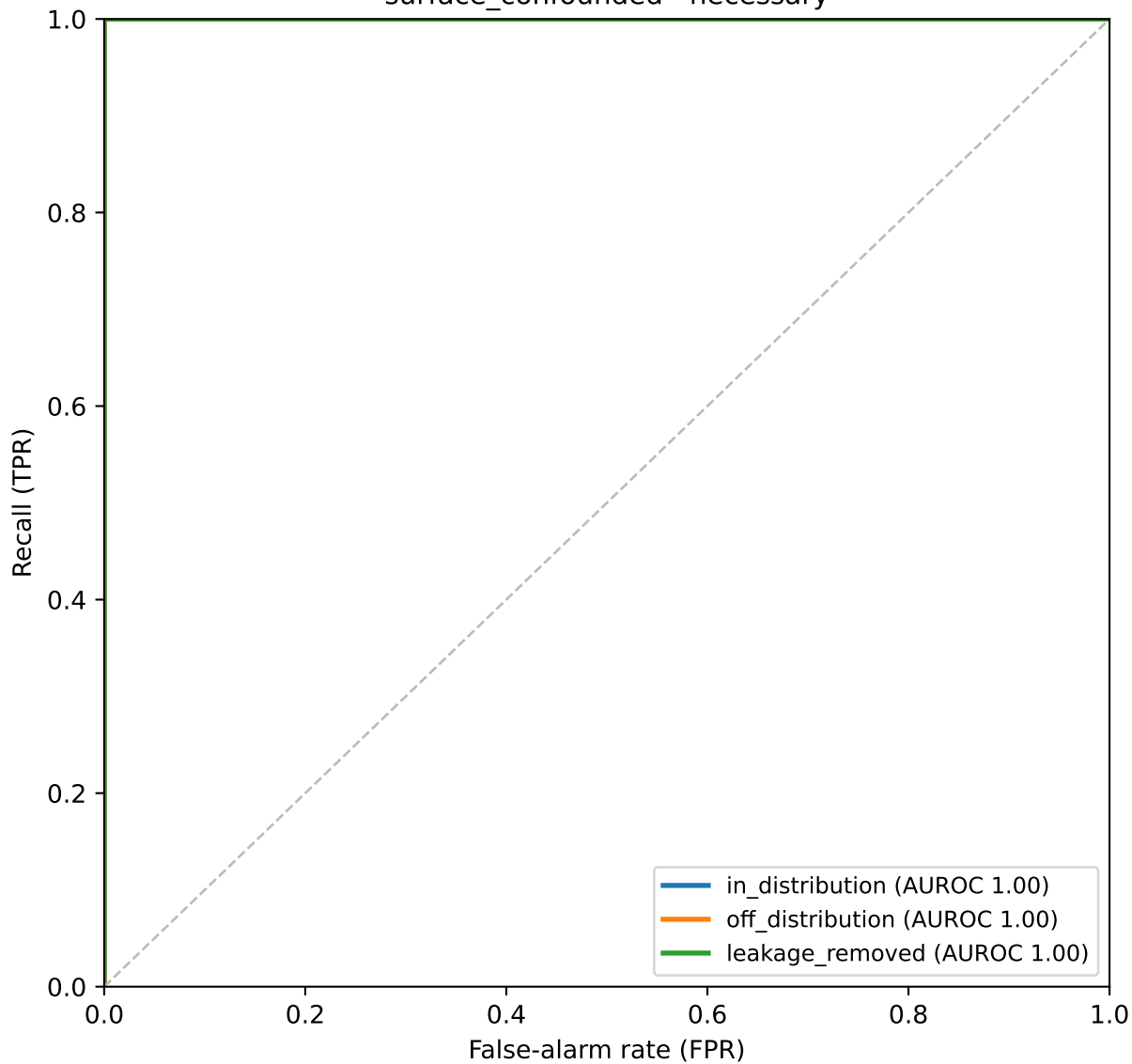


SIEVE deployment ROC
surface_confounded · necessary



SIEVE deployment report

Verdict: surface_confounded · necessary

What this means for a deployer:

In-distribution, at a 5% false-alarm rate this probe flags about 100% of the cases it should (95% CI 100%-100%).

Off-distribution (a different prompt mix, closer to the wild), the same 5% false-alarm setting catches about 100% (95% CI 100%-100%).

With the giveaway text removed, recall at 5% FPR falls to about 100%: the realistic floor when the behavior isn't spelled out in the transcript.

Bottom line: choose your threshold from the ROC curve — lower it to miss fewer cases (more false alarms), raise it to cut false alarms (more misses).

Condition	FPR budget	Recall (95% CI)
in_distribution	1%	100% [100, 100]
in_distribution	5%	100% [100, 100]
in_distribution	10%	100% [100, 100]
off_distribution	1%	100% [100, 100]
off_distribution	5%	100% [100, 100]
off_distribution	10%	100% [100, 100]
leakage_removed	1%	100% [100, 100]
leakage_removed	5%	100% [100, 100]
leakage_removed	10%	100% [100, 100]